

Move■In by PayServe

Comprehensive System Architecture & Platform Overview

This document provides a detailed overview of the Move■In by PayServe platform. It summarizes the architecture, technology stack, domain modules, API structure, data model, scalability strategy, and security architecture designed to support a modern trusted rental marketplace.

Core Vision

Move■In is designed to digitize the entire rental lifecycle — from discovery and viewing to application, screening, reservation, payment, and move■in. The platform focuses on transparency, verification, and structured rental processes to improve trust between tenants, landlords, caretakers, and administrators.

1. Platform Actors

- Tenants – search for homes, book viewings, submit rental applications, reserve homes.
- Landlords – list properties, manage units, review applications, approve reservations.
- Caretakers – assist in verification, coordinate viewings, provide property updates.
- Administrators – maintain platform trust through moderation and compliance.

2. Technology Stack

- Frontend: React.js single-page application delivering responsive user interfaces.
- Backend: Node.js with Fastify providing high-performance APIs and modular architecture.
- Database: MongoDB document database supporting flexible schema and scalable data storage.
- Caching & Jobs: Redis and BullMQ for background job processing.
- Media Storage: Object storage for images, videos, and verification documents.
- Infrastructure: Load balanced servers, CDN edge protection, and monitoring systems.

3. System Architecture Layers

- Client Layer – tenant portal, landlord dashboard, caretaker interface, and admin console.
- Edge Layer – CDN, Web Application Firewall, TLS termination, and rate limiting.
- Application Layer – Fastify API services handling validation, authentication, and logic.
- Domain Layer – modular business services such as listings, viewing scheduler, and payments.
- Data Layer – MongoDB cluster, Redis cache, and background workers.

4. Core Domain Modules

- Identity & Authentication – registration, login, token management, role-based access.
- User Profiles – personal information, preferences, saved listings.
- Properties & Units – physical building and rental unit management.
- Listings – public marketplace representation of rental units.
- Verification – landlord, unit, and caretaker verification processes.
- Viewing Scheduler – booking and managing property viewings.
- Applications – structured tenant rental applications.
- Screening – review and risk assessment of applications.
- Reservations – temporary unit holding before final move-in.
- Payments – deposits, booking fees, and other financial transactions.
- Messaging – tenant, landlord, and caretaker communication.
- Notifications – in-app, SMS, and email alerts.
- Admin & Moderation – trust and safety oversight.

5. Database Architecture

MongoDB stores all operational data using structured collections. The design separates core rental concepts to maintain data integrity and scalability.

- users
- landlords
- tenants
- caretakers
- properties
- units
- listings
- verification_records
- viewing_slots
- viewing_bookings
- applications
- screening_reviews
- reservations
- payments
- conversations
- messages
- notifications
- audit_logs

6. Security Architecture

- Encrypted HTTPS communication across all platform interactions.
- Token-based authentication and refresh token rotation.
- Role-based access control combined with ownership policy checks.
- Input validation and schema enforcement using Fastify.
- Secure file upload and document storage.
- Audit logging for sensitive actions.
- Secrets and credentials stored securely outside application code.

7. Scalability Strategy

- Horizontal scaling of backend servers behind load balancers.
- Stateless API design allowing independent server scaling.
- MongoDB replica sets for high availability.
- Redis caching to reduce database load.
- Asynchronous job processing using BullMQ workers.

8. MVP vs Full Platform

The platform will be developed in phases. The MVP focuses on validating the rental funnel while the full platform expands to property management and tenant lifecycle services.

Phase	Capabilities
MVP	Listings, search, viewing scheduling, applications, reservations, payments, messaging, a
Full Platform	Lease management, rent collection automation, maintenance workflows, analytics dashb

MoveIn by PayServe is designed to evolve into a complete digital infrastructure for the residential rental ecosystem — enabling safer transactions, structured property management, and trusted tenant-landlord relationships.